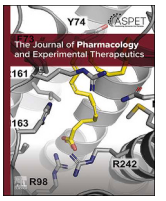
The Journal of Pharmacology and Experimental Therapeutics 392 (2025) 103720

The Journal of Pharmacology and Experimental Therapeutics

journal homepage: www.jpet.aspetjournals.org



ARTICLE

Mitragynine and 7-hydroxymitragynine: Bidirectional effects on breathing in rats

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ARTICLE INFO

Article history:
Received 14 May 2025
Accepted 16 September 2025
Available online 22 September 2025

Key words:
Kratom
Mitragynine
7-Hydroxymitragyine
Respiratory depression
Opioid overdose

ABSTRACT

The use of kratom as an alternative to conventional opioids has surged, driven largely by anecdotal reports of its efficacy for pain relief and opioid withdrawal management. The growing prevalence of kratom products enriched with 7-hydroxymitragynine (7-HMG), an active metabolite of mitragynine (MG), necessitates evaluating the respiratory effects of these alkaloids and determining whether naloxone reverses their potential respiratory depressant effects. Respiratory parameters were measured in awake, freely moving female and male Sprague-Dawley rats using whole body plethysmography. To minimize handling-induced artifacts and ensure precise respiratory recordings, drugs were administered intravenously. Morphine and 7-HMG induced significant respiratory depression, evidenced by reductions in breathing frequency, tidal volume, and minute volume. The potency of 7-HMG to decrease minute volume by 50% was 4.5-fold greater than that of morphine. In contrast, MG administration unexpectedly increased respiratory frequency. Naloxone fully reversed the respiratory depression induced by both morphine and 7-HMG but did not alter the respiratory stimulant effects produced by MG. These findings demonstrate that 7-HMG exhibits significant respiratory depressant properties similar to classical opioids, and importantly, such depressant effects are effectively antagonized by naloxone. Conversely, MG exerts respiratory stimulant effects through mechanisms independent of opioid receptor pathways. Collectively, these data highlight crucial pharmacological distinctions between kratom alkaloids, underscoring the risk associated with high 7-HMG–containing kratom products and suggesting that the predominant alkaloid MG may offer a safer respiratory profile.

Significance Statement: The prevalence of kratom products containing 7-hydroxymitragynine (7-HMG), a μ-opioid receptor agonist, underscores the need to evaluate respiratory effects of kratom-related alkaloids and their reversal by naloxone. 7-HMG induced significant respiratory depression comparable with morphine, which was reversed by naloxone. Conversely, mitragynine, kratom's most abundant alkaloid, unexpectedly increased respiratory frequency unaffected by naloxone. These findings highlight critical pharmacological differences between kratom-related alkaloids, emphasizing potential risks associated with products containing high concentrations of 7-HMG.

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1. Introduction

Opioids have long been recognized for their analgesic properties, yet their clinical utility is hindered by their potentially life-threatening side effects, like respiratory depression.[1–3] Opioid-induced respiratory depression (OIRD) represents a significant risk in both clinical settings and abuse-related overdoses.[4,5] Preclinical

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This article has supplemental material available at jpet.aspetjournals.org.

https://doi.org/10.1016/j.jpet.2025.103720
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studies have elucidated the mechanisms underlying OIRD, implicating μ-opioid receptor (MOR) activation in brainstem respiratory centers.[6,7] Clinical research has further highlighted the risk of OIRD in vulnerable populations such as in older adults and those with sleep-disordered breathing.[8,9]

These safety concerns have driven the search for alternative analgesics with improved respiratory profiles. Kratom (*Mitragyna speciosa*) has garnered increasing attention for its opioid-like effects and potential therapeutic applications. In recent years, the use of kratom as an alternative to conventional opioids within the public domain has surged, driven by anecdotal reports of efficacy in managing pain and mitigating opioid withdrawal symptoms.[10–12] Beyond these anecdotal reports, scientific interest in kratom continues to grow, with several studies focusing on its primary alkaloid, mitragynine (MG), and its hydroxylated metabolite, 7-hydroxymitragynine (7-HMG).

Although both MG and 7-HMG function as biased agonists at the MOR,[13–16] such biased agonism does not necessarily confer improved safety over conventional opioids. The concept of biased agonism, which involves selective activation of specific signaling pathways over others, was initially expected to provide a path to safer opioid medications by preferentially activating analgesic pathways while avoiding those associated with adverse effects. However, compounds developed as biased MOR agonists have failed to demonstrate improved safety profiles.[17–20] This disconnect highlights the complexity of MOR signaling and suggests that biased agonism alone may not be sufficient to overcome the fundamental challenges of opioid pharmacology.

Pharmacological characterization of the abuse liability of MG and 7-HMG revealed striking differences. MG did not substitute for morphine in the drug discrimination assay and was not self-administered, whereas 7-HMG fully substituted for morphine and demonstrated reinforcing effects.[21–24] Additionally, 7-HMG in vivo has been demonstrated to be 3- to 13-fold more potent than morphine.[24–26] These findings suggest that 7-HMG, but not MG, has greater liability risk and abuse potential similar to other opioid agonists despite biased agonism.

The differential abuse potential between these alkaloids is particularly concerning given the current market trends. A significant concern is the increasing availability of kratom products that contain high concentrations of 7-HMG.[27,28] Although analyses indicate that 7-HMG is either not detectable or present in extremely small amounts (<0.1%) in the living plant,[29,30] likely formed as a postharvest artifact,[31] some manufacturers are marketing products containing 14–25 mg of 7-HMG per recommended serving, with labeled purities as high as 98%.[28] This poses a risk, particularly for kratom-naïve individuals who may unknowingly purchase these products under the assumption that they are typical kratom preparations. These concerns underscore the need to evaluate the safety of both 7-HMG and MG, particularly regarding their effects on respiration.

The current state of knowledge regarding the respiratory effects of kratom and its related alkaloids remains limited and somewhat conflicting. One study found that oral MG did not yield significant changes in blood gases in rats, even at high doses[32] though this study measured blood gases rather than direct respiratory parameters. Another study in mice reported respiratory depression but observed a ceiling effect in MG-induced respiratory depression, whereas 7-HMG showed dose-dependent effects.[33] However, it is important to note that mice exhibit hyperlocomotion following MOR agonist administration,[34–38] potentially making rats the more appropriate species for studying OIRD. Additionally, the naloxone-reversibility of these respiratory effects remains to be determined. This is crucial as it would inform clinicians whether naloxone is effective in reversing 7-HMG induced respiratory depression.

In this study, we investigated the respiratory effects of the prototypical opioid, morphine, and kratom-related alkaloids (MG and 7-HMG) using whole body plethysmography in rats in 2 distinct experimental phases. In the first phase, we evaluated the direct respiratory effects of morphine, MG, and 7-HMG. In the second phase, we examined the potential for naloxone as a rescue intervention. This investigation of the respiratory effects and naloxone-reversibility of these compounds provides new insights into the role of opioid receptor mechanisms in kratom-induced respiratory depression.

2. Material and methods

2.1. Subjects

A total of 32 Sprague-Dawley rats (Charles River; $n = 16$ females, $n = 16$ males), aged 12 weeks, were used in this study. The rats were purchased with preimplanted indwelling jugular catheters. For the first experimental phase, 16 animals were divided into 2 groups ($n = 8$, morphine group; $n = 8$, MG/7-HMG group). All testing followed a repeated-measures (RM), within-subjects design where each animal received multiple doses of their respective drug(s). One male rat from the kratom-related alkaloids group expired before the completion of the experiment owing to postsurgical complications, resulting in a final sample size of $n = 7$ for this group. The remaining 16 animals were used in the second experimental phase, with 8 animals assigned to the morphine–naloxone group and 8 animals assigned to the kratom-related alkaloids–naloxone group. Animals were housed individually within a temperature- and humidity-controlled vivarium and kept on a 12:12 light-dark cycle (lights on at 07:00 AM). Food and water were provided ad libitum. This study was conducted in accordance with the Guide for the Care and Use of Laboratory Animals[39] and was approved by the Institutional Animal Care Use Committee at Texas Tech University Health Sciences Center.

2.2. Drugs

Morphine sulfate and naloxone hydrochloride (Fagron) were diluted in normal saline. Mitragynine (MG) HCl and 7-hydroxymitragynine (7-HMG) were isolated as previously described[40] and diluted in a vehicle solution consisting of 2.5% Tween 20, 2.5% Tween 80, 25% polyethylene glycol, 25% propylene glycol, and 45% normal saline. Heparin sodium salt from porcine intestinal mucosa (MilliporeSigma), gentamicin (Covetrus), ketamine (Covetrus), and midazolam (Zoetis) were all diluted in normal saline.

2.3. Apparatus

A whole body plethysmography system (Data Sciences International) was used to measure and record the respiratory parameters in awake, freely moving rats. The system consisted of 8 cylindrical acrylic chambers measuring 18.4 cm in diameter and 13.1 cm in height. The top interior portion of the chamber houses an infusion tether that is connected magnetically to the implanted vascular access button on the rats. The exterior portion of the chamber is connected to a pinport that facilitates aseptic intravenous drug delivery (PNP3M; Instech laboratories). The intravenous route was chosen to eliminate handling-induced respiratory artifacts during drug delivery, ensuring undisturbed respiratory recordings while enabling maximal effects with lower drug doses. Similarly, white noise was continuously played in the testing room

to mask external sounds and minimize respiratory disturbances that could be caused by environmental noises.

The whole body plethysmography system measures box flow (milliliters per second), which refers to the airflow in and out of the chamber caused by the animal's respiration. Changes in air temperature and humidity during the breathing cycle primarily drive pressure/box flow changes recorded using whole body plethysmography.[41] When an animal inhales, the air is warmed and humidified. When it exhales the warmed air, the air temperature and humidity in the chamber increase, and pressure subsequently increases. Therefore, exhalations are recorded as upward deflections in pressure/box flow, and inhalations are recorded as downward deflections in pressure/box flow. These pressure/box flow changes are detected by a pressure transducer and processed by the whole body plethysmography software Finepointe v.3.1 (DSI) to calculate various respiratory parameters.

Respiratory rate (breaths per minute) is directly measured from fluctuations in the positive and negative box flow. Tidal volume (milliliters per breath) and minute volume (milliliters per minute) are estimated from the box flow signal. Tidal volume is equal to the product of the volume measured from the box flow and a compensation factor calculated using the formula by Drorbaugh and Fenn,[42] which accounts for temperature, humidity, and barometric pressure. Minute volume is equal to the product of respiratory rate and tidal volume (rate $\times$ box flow volume $\times$ compensation factor). In addition to these parameters, the system also calculates apneas. An apnea is defined as an unusually long breath (at least 1 second long and 100% greater than the breaths 5 minutes before).

2.4. Study design

The study was conducted in 2 distinct experimental phases. In the first phase, we evaluated the direct respiratory effects by administering either saline or morphine (10 and 32 mg/kg, i.v.) in 1 group ($n = 8$). A separate group ($n = 7$) received either vehicle or the kratom-related alkaloids MG (5.6 and 10 mg/kg, i.v.) and 7-HMG (1, 3.2, and 10 mg/kg, i.v.). The order of drug administration was counterbalanced between subjects in both groups. For MG, we observed stimulatory effects but no respiratory depression at doses up to 10 mg/kg. On the basis of these findings, all animals in the kratom alkaloid group received a higher 17.8 mg/kg (i.v.) MG dose to assess potential respiratory depression at higher doses. This highest dose still produced no significant respiratory depression and resulted in seizure-like activity in 2 of 3 male subjects, with 1 animal sustaining physical injury to the digits of its hind paw. This adverse reaction established 10 mg/kg as our upper safety limit for subsequent testing.

In the second phase, using entirely different animals, we examined the extent to which the opioid antagonist naloxone could reverse respiratory effects. The doses used in this phase (32 mg/kg for morphine, 10 mg/kg for MG, and 10 mg/kg for 7-HMG) were selected on the basis of their significant effects on respiratory parameters during the first phase. One cohort ($n = 8$) underwent 4 treatment conditions in a fully counterbalanced Latin square design: saline followed by saline, saline followed by naloxone (1 mg/kg, i.v.), morphine (32 mg/kg, i.v.) followed by saline, and morphine (32 mg/kg, i.v.) followed by naloxone (1 mg/kg, i.v.). Another cohort ($n = 8$) underwent 6 treatment conditions using Williams design: vehicle followed by saline, vehicle followed by naloxone (1 mg/kg, i.v.), MG (10 mg/kg, i.v.) followed by saline, MG (10 mg/kg, i.v.) followed by naloxone (1 mg/kg, i.v.), 7-HMG (10 mg/kg, i.v.) followed by saline, and 7-HMG (10 mg/kg, i.v.) followed by naloxone (1 mg/kg, i.v.). In all second-phase treatments, naloxone or saline was administered 10 minutes after the initial compound injection.

Dose selection for this study was determined through preliminary testing in our laboratory. For morphine, doses of 10 and 32 mg/kg (i.v.) were selected on the basis of pilot studies demonstrating that higher doses were necessary to produce reliable respiratory depression under normal air conditions without CO_2 challenge, which typically reveals respiratory depression at lower doses. For the kratom-related alkaloids, we implemented a systematic dose-escalation approach due to the absence of published intravenous administration data. For the naloxone antagonism studies, we selected a dose of 1 mg/kg (i.v.) to provide effective blockade of opioid receptor-mediated effects given the relatively high doses of the test compounds used in our paradigm.

Animals were given a 1-week habituation to new housing conditions before experiments commenced. After habituation, rats were transported to an experimental room mimicking conditions of the housing room and were placed individually in a whole body plethysmography chamber for a 3-hour acclimation period. During this time, rats received 2 vehicle injections to familiarize them with the injection procedure before returning to the housing room. For experimental sessions, each weekly testing began with a 60-minute acclimation period followed by a 20-minute baseline recording. After delivery of test compounds, respiratory parameters were recorded continuously for 60 minutes in both phases. Injections were manually administered over 15–30 seconds, and infusion lines were flushed with saline to ensure complete drug delivery.

To maintain catheter patency and prevent infections or blood clots, catheters were flushed with a solution containing gentamicin (4 mg/mL) and heparin (30 IU/mL) following catheter manipulation and at least every 3 days.[43] On the day after each test session (Fridays), catheter patency was verified by infusing 0.2 mL of a solution containing ketamine (15 mg/mL) and midazolam (0.5 mg/mL) into the catheter.[44] Loss of righting reflex within 5 seconds indicated a patent catheter.

2.5. Statistical analysis

The study timeline consisted of 60 minutes of acclimation to whole body plethysmography chambers, followed by a 20-minute baseline recording period. The acclimation period was neither plotted nor included in any analyses. After baseline recording, test compounds were administered intravenously, and postdrug measurements continued for 60 minutes, with data collected in 5-minute bins. To account for individual differences and associated baseline measures, all respiratory parameters (frequency, tidal volume, and minute volume) were normalized by expressing each postdrug value as a percentage of the corresponding predrug baseline value. This normalization is particularly important for tidal volume, which is directly influenced by mass across species.[45–48] For each experimental session, baseline values were calculated as the average of the 20-minute recording period before drug infusion. Raw trace data for significant doses of each compound are shown in Supplemental Figs. 1–4, and the average raw frequency data are shown in Supplemental Table 1. For quantitative analysis, the time course data were transformed to area under the curve (AUC) values using the trapezoidal method, calculated over 2 specific time intervals: 0–30 minutes and 0–60 minutes postadministration.

For single-drug experiments with morphine, MG, and 7-HMG, AUC values were analyzed using 1-way RM ANOVAs separately for each respiratory parameter at the 30- and 60-minute intervals. Greenhouse–Geisser correction was applied to adjust degrees of freedom when the assumption of sphericity was violated.

Following significant main effects, Dunnett's multiple comparisons test was used to compare drug treatment groups against the vehicle control. ED_{50} values were calculated using simple linear regression if a drug decreased minute volume below 50% of baseline.

In the naloxone antagonism experiments, morphine, MG, or 7-HMG administration was followed by naloxone or saline 10 minutes later. AUC values were calculated as described earlier and analyzed using 2-way RM ANOVAs with naloxone treatment and drug treatment as factors. The analysis examined the main effects of naloxone treatment and drug treatment, and their interaction at both time intervals. Tukey multiple comparisons test was used for multiple comparisons between treatment conditions. Sex differences, ie, differences between male and female rats, were analyzed using 2-way RM ANOVA (Supplemental Table 2), and no significant differences were observed. Statistical significance was established at $P < .05$ for all analyses. All statistical procedures were performed using GraphPad Prism (version 10.3.1) software.

3. Results

3.1. Respiratory effects of morphine, MG, and 7-HMG alone

Morphine produced dose-dependent respiratory depression, as illustrated by time courses for respiratory frequency, tidal volume, and minute volume (Fig. 1A–C). Representative raw trace data for saline control and morphine are shown in Supplemental Figs. 1 and 2, respectively. Morphine had no significant effect on respiratory frequency at either 30 minutes (Fig. 1D) ($F[1.57,11.0] = 3.65$; $P = .069$) or 60 minutes (Fig. 1E) ($F[1.58,11.1] = 3.06$; $P = .096$). Morphine significantly affected tidal volume at both 30 minutes (Fig. 1F) ($F[1.25,8.76] = 10.9$; $P = .007$) and 60 minutes (Fig. 1G) ($F[1.35,9.42] = 16.9$; $P = .001$). Post hoc analysis revealed that only the higher dose of morphine (32 mg/kg) significantly reduced tidal volume at both time points ($P < .001$). Similarly, morphine significantly decreased minute volume at both 30 minutes (Fig. 1H) ($F[1.31,9.15] = 8.79$; $P = .012$) and 60

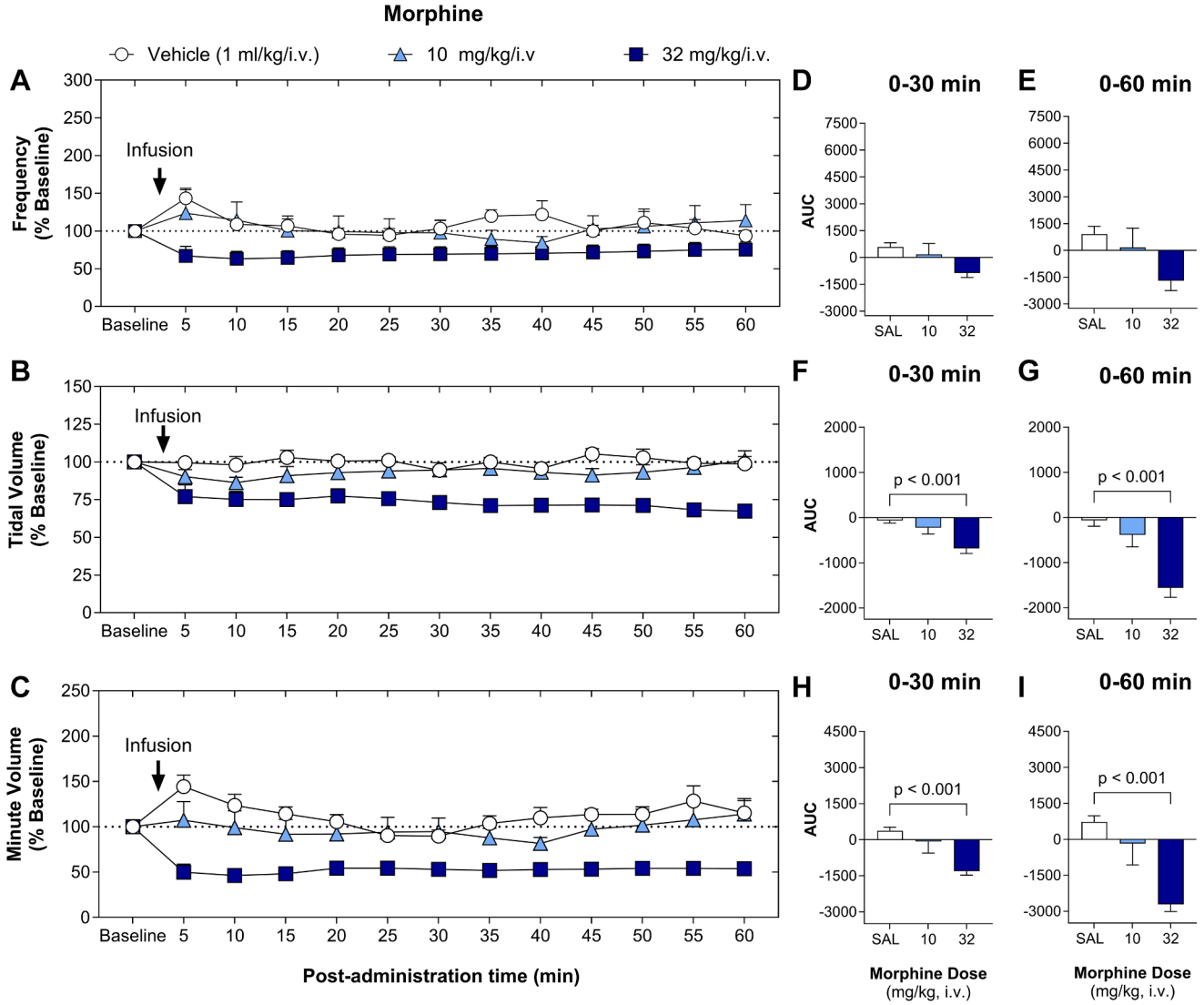


Fig. 1. Time-course and AUC of morphine effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels), following intravenous administration of morphine in rats. Morphine was administered at doses of 10 and 32 mg/kg, with saline (1 mL/kg) as the control. Data points for the time course (A–C) represent the mean ± SEM ($n = 8$), recorded in 5-minute bins continuously for up to 60 minutes postinfusion. Data for AUC are shown as mean ± SEM ($n = 8$) for 30 (D, F, and H) and 60 (E, G, and I) minutes postinfusion. Statistically significant differences between saline and different doses are indicated (P values). The ED_{50} for morphine to decrease minute volume by 50% was 31.5 mg/kg (24.0–47.8 mg/kg).

minutes (Fig. 1I) (F[1.21,8.44] = 9.73; $P = .011$). Post hoc analysis showed that only the higher dose of morphine (32 mg/kg) significantly reduced minute volume at both time points ($P < .001$), whereas the lower dose (10 mg/kg) did not produce significant respiratory depression. The ED_{50} for morphine to decrease minute volume by 50% was 31.5 mg/kg (24.0–47.8 mg/kg). Quantification of apneas during the first 15 minutes showed a nonsignificant trend for the highest dose of morphine (Supplemental Fig. 5).

MG administration produced respiratory effects distinct from morphine, as shown in the time courses for frequency, tidal volume, and minute volume (Fig. 2A–C). Representative raw trace data for significant MG doses are shown in Supplemental Fig. 3. MG significantly increased respiratory frequency at both 30 minutes (Fig. 2D) (F[1.99,11.9] = 4.00; $P = .047$) and 60 minutes (Fig. 2E) (F[1.78,10.7] = 4.77; $P = .036$). Post hoc analysis revealed that the 10-mg/kg dose significantly elevated respiratory frequency compared with vehicle control at both 30 minutes ($P = .035$) and 60 minutes ($P = .046$). MG did not significantly affect tidal volume at either 30 minutes (Fig. 2F) (F[1.53,9.20] = 3.40; $P = .086$) or 60 minutes (Fig. 2G) (F[1.35,8.09] = 3.93; $P = .075$). Similarly, minute volume was not significantly altered by MG administration at either 30 minutes (Fig. 2H) (F[1.53,9.17] = 4.07; $P = .062$) or 60 minutes (Fig. 2I) (F[1.59,9.53] = 2.65; $P = .128$).

Administration of 7-HMG produced robust, dose-dependent respiratory depression. Time courses for respiratory frequency, tidal volume, and minute volume are presented in Fig. 3A–C. Representative raw trace data for significant 7-HMG doses are shown in Supplemental Fig. 4. Respiratory frequency was significantly decreased by 7-HMG at both 30 minutes (Fig. 3D) (F[1.91,11.5] = 12.7; $P = .001$) and 60 minutes (Fig. 3E) (F[2.26,13.6] = 7.72; $P = .005$). Post hoc analysis revealed that both 3.2- and 10-mg/kg doses significantly reduced respiratory frequency at 30 minutes ($P = .04$ and $P = .03$, respectively), but these effects were not maintained at 60 minutes. For tidal volume, 7-HMG approached but did not reach significance at both 30 minutes (Fig. 3F) (F[1.55,9.31] = 4.19; $P = .058$) and 60 minutes (Fig. 3G) (F[1.57,9.40] = 4.05; $P = .061$). Minute volume was significantly

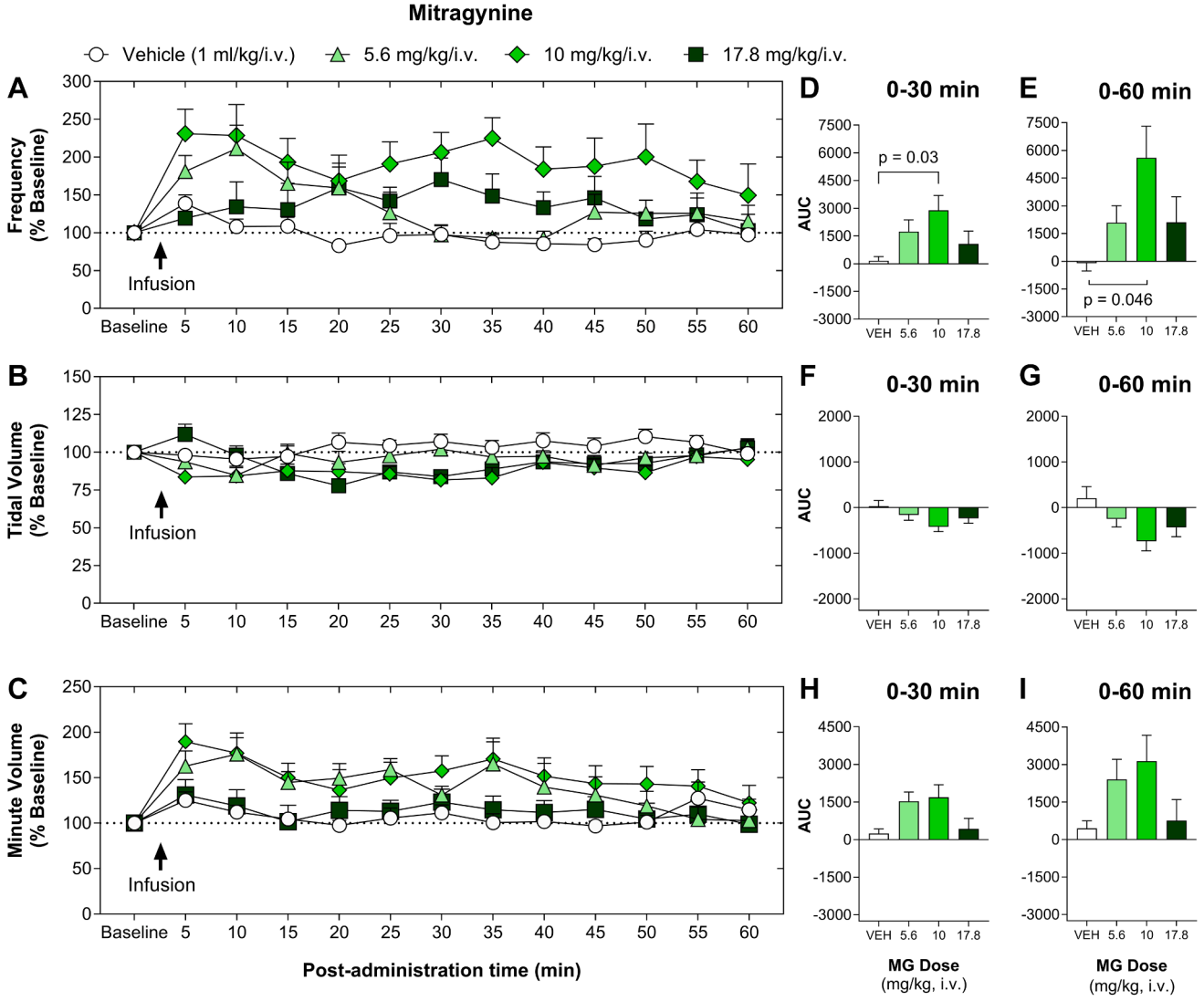


Fig. 2. Time-course and AUC of MG effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels) following intravenous administration of MG in rats. MG was administered at doses of 5.6, 10, and 17.8 mg/kg, with vehicle (1 mL/kg) as the control. Data points for the time course (A–C) represent the mean ± SEM (n = 7), recorded in 5-minute bins continuously for up to 60 minutes postinfusion. Data for AUC are shown as mean ± SEM (n = 7) for 30 (D, F, and H) and 60 (E, G, and I) minutes postinfusion. Statistically significant differences between vehicle and different doses are indicated (P values).

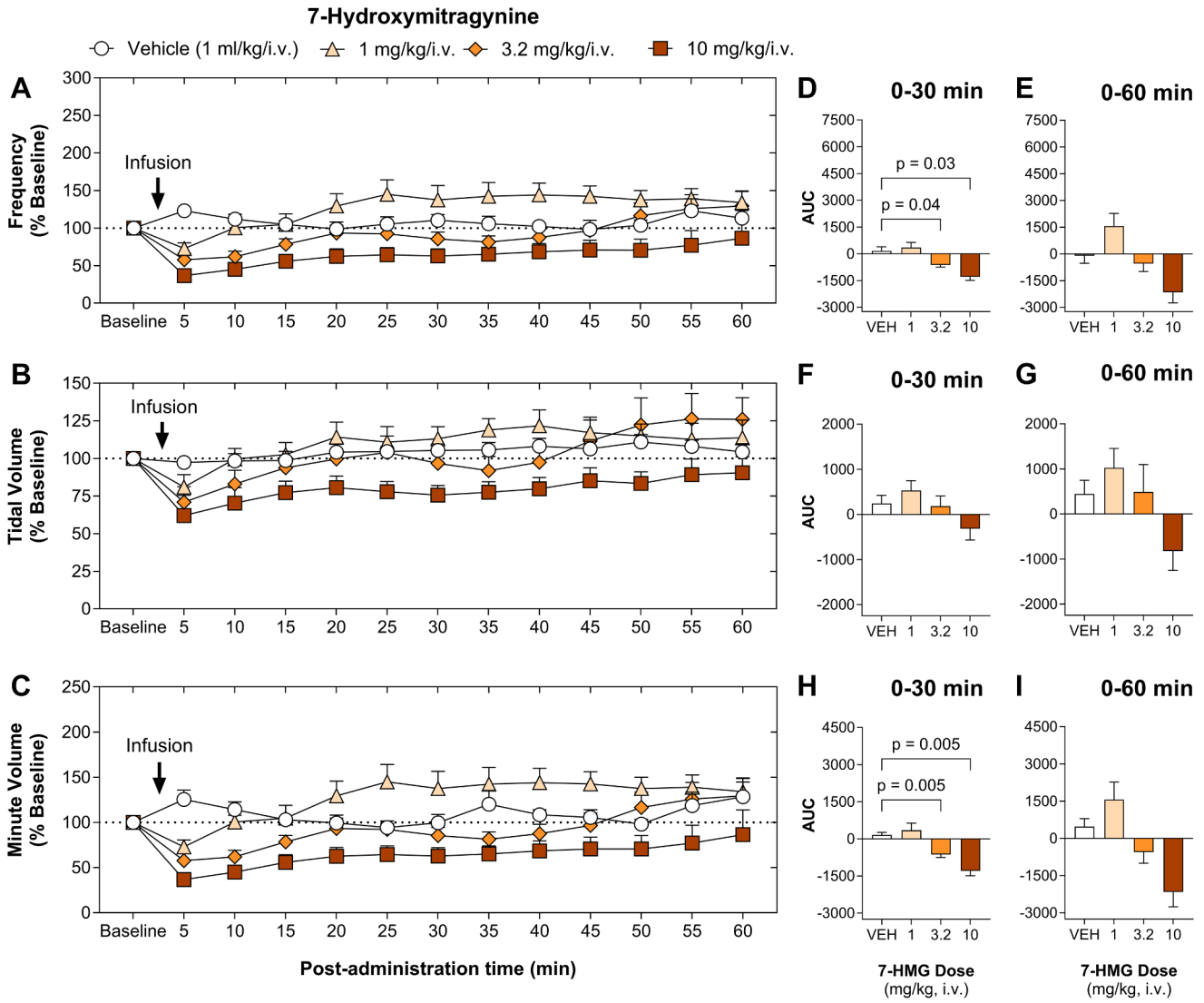


Fig. 3. Time course and AUC of 7-HMG effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels) following intravenous administration of 7-HMG in rats. 7-HMG was administered at doses of 1, 3.2, and 10 mg/kg, with vehicle (1 mL/kg) as the control. Data points for the time course (A–C) represent the mean ± SEM ($n = 7$), recorded in 5-minute bins continuously for up to 60 minutes postinfusion. Data for AUC are shown as mean ± SEM ($n = 8$) for 30 (D, F, and H) and 60 (E, G, and I) minutes postinfusion. Statistically significant differences between vehicle and different doses are indicated (P values). The ED_{50} for 7-HMG to decrease minute volume by 50% was 6.98 mg/kg (5.20–10.23 mg/kg).

reduced by 7-HMG at both 30 minutes (Fig. 3H) ($F[2.17,13.0] = 17.9$; $P < .001$) and 60 minutes (Fig. 3I) ($F[2.35,14.1] = 8.52$; $P = .003$). Post hoc analysis revealed that the highest doses (3.2 and 10 mg/kg/i.v.) significantly decreased minute volume at 30 minutes ($P = .005$ for both doses), but these effects were not statistically significant at 60 minutes. The ED_{50} for 7-HMG to decrease minute volume by 50% was 6.98 mg/kg (5.20–10.23 mg/kg). Quantification of apneas during the first 15 minutes revealed a significant effect for the highest dose of 7-HMG (Supplemental Fig. 5).

3.2. Naloxone antagonism studies

3.2.1. Morphine + naloxone

Naloxone (1 mg/kg) treatment effectively reversed the respiratory depression induced by morphine (32 mg/kg). Time courses for respiratory frequency, tidal volume, and minute volume are presented in Fig. 4A–C. For respiratory frequency, 2-way RM ANOVA at 30 minutes revealed nonsignificant main effects of naloxone ($F[1,7] = 3.74$; $P = .09$) or morphine ($F[1,7] = 4.71$; $P = .07$), but a significant naloxone × morphine interaction ($F[1,7] = 15.5$; $P = .009$) (Fig. 4D). Multiple comparisons tests revealed that neither morphine + saline nor morphine + naloxone administration significantly impacted respiratory frequency compared with control, consistent with the results of the morphine dose-response experiment. At 60 minutes, these effects accumulate with significant main effects of naloxone ($F[1,7] = 7.09$; $P = .03$), morphine ($F[1,7] = 7.58$; $P = .03$), and naloxone × morphine interaction ($F[1,7] = 18.1$; $P = .004$) (Fig. 4E). At this time point, morphine + naloxone was significantly different from morphine + saline ($P = .04$), demonstrating an overshoot effect.

For tidal volume at 30 minutes, a significant effect of morphine ($F[1,7] = 34.2$; $P < .001$) was observed, whereas naloxone ($F[1,7] = 1.99$; $P = .20$) and naloxone × morphine interaction ($F[1,7] = 0.185$; $P = .68$) did not reach significance (Fig. 4F). At 60 minutes, both naloxone ($F[1,7] = 6.52$; $P = .04$) and morphine ($F[1,7] = 34.5$; $P < .001$) showed significant effects but not the naloxone × morphine

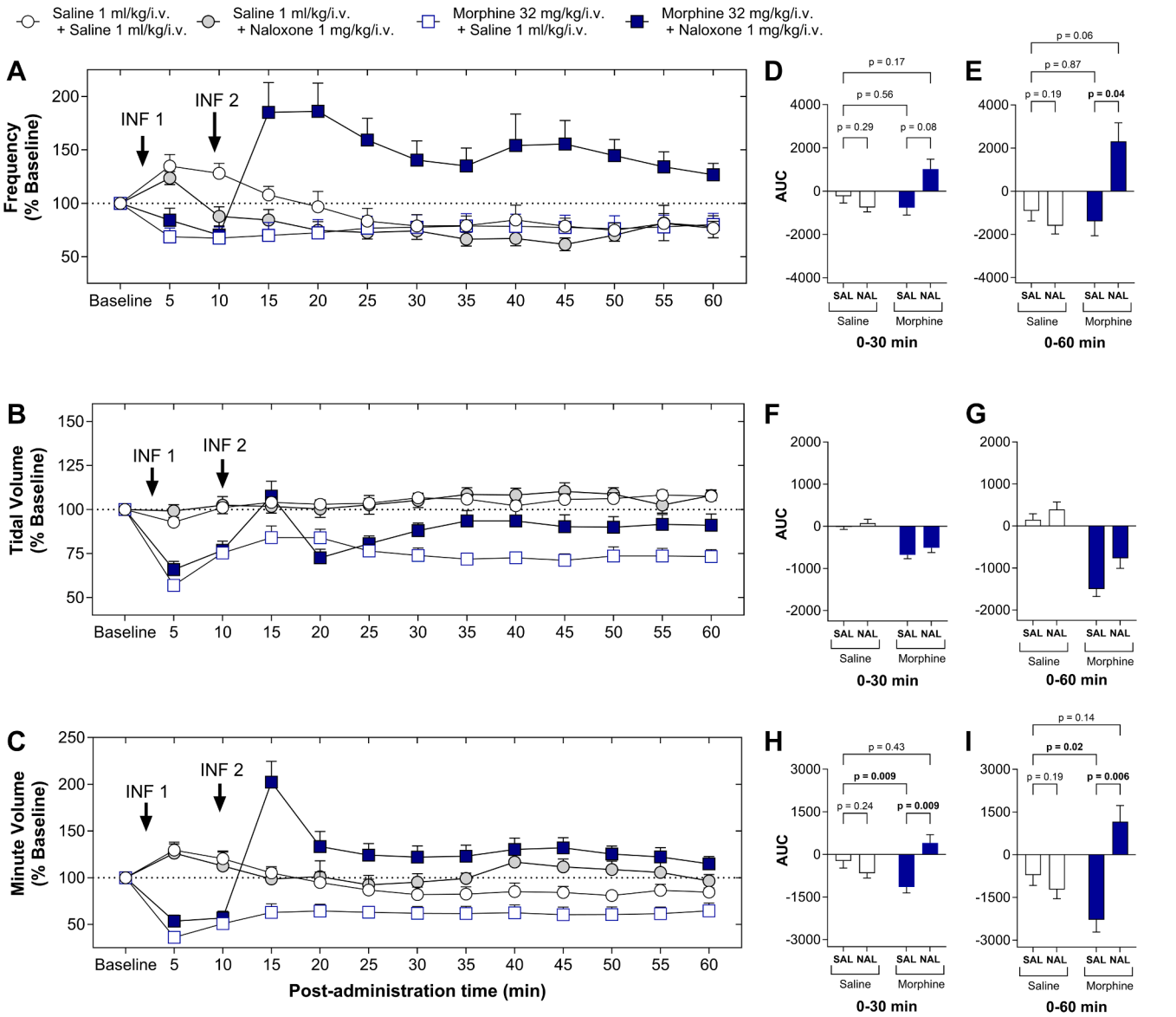


Fig. 4. Time course and AUC of morphine and naloxone effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels) following intravenous administration of morphine with naloxone rescue treatment in rats. Morphine was administered (32 mg/kg), followed by naloxone (1 mg/kg) rescue treatments after 10 minutes. Data points for the time course (A–C) represent the mean ± SEM ($n = 7$), recorded in 5-minute bins continuously for up to 60 minutes postmorphine infusion. Data for AUC are shown as mean ± SEM ($n = 8$) for 30 (D, F, and H) and 60 (E, G, and I) minutes postmorphine infusion. Statistically significant differences between treatments are indicated (P values).

interaction ($F[1,7] = 2.07$; $P = .19$) (Fig. 4G). No post hoc analyses were conducted for tidal volume owing to the absence of significant interactions at either time point.

For minute volume at 30 minutes, 2-way RM ANOVA showed significant effects of naloxone ($F[1,7] = 6.46$; $P = .04$) and naloxone × morphine interaction ($F[1,7] = 36.5$; $P < .001$) (Fig. 4H). Post hoc analysis revealed significant depression with morphine alone compared with that with control ($P = .009$), which was not significant when naloxone was coadministered. Additionally, morphine + saline and morphine + naloxone treatments were significantly different from each other ($P = .009$). At 60 minutes, significant effects of naloxone ($F[1,7] = 13.1$; $P = .008$) and naloxone × morphine interaction ($F[1,7] = 40.2$; $P < .001$) persisted (Fig. 4I). The same pattern of post hoc differences was observed at this time point, with morphine alone causing significant depression compared with control ($P = .02$), whereas morphine + naloxone was not significantly different from control but was significantly different from the morphine + saline group ($P = .006$).

3.2.2. MG + naloxone

Administration of naloxone failed to antagonize MG's respiratory effects. Time courses for respiratory frequency, tidal volume, and minute volume are presented in Fig. 5A–C, respectively. Two-way RM ANOVA for respiratory frequency showed that MG had a significant effect at both 30 (Fig. 5D) ($F[1,7] = 14.4$; $P = .007$) and 60 minutes (Fig. 5E) ($F[1,7] = 8.99$; $P = .020$), whereas naloxone effects and interactions were not significant at either time point. Despite the lack of naloxone effects, the significant main effect of MG indicates that both MG + saline and MG + naloxone treatments resulted in elevated respiratory frequencies compared with controls.

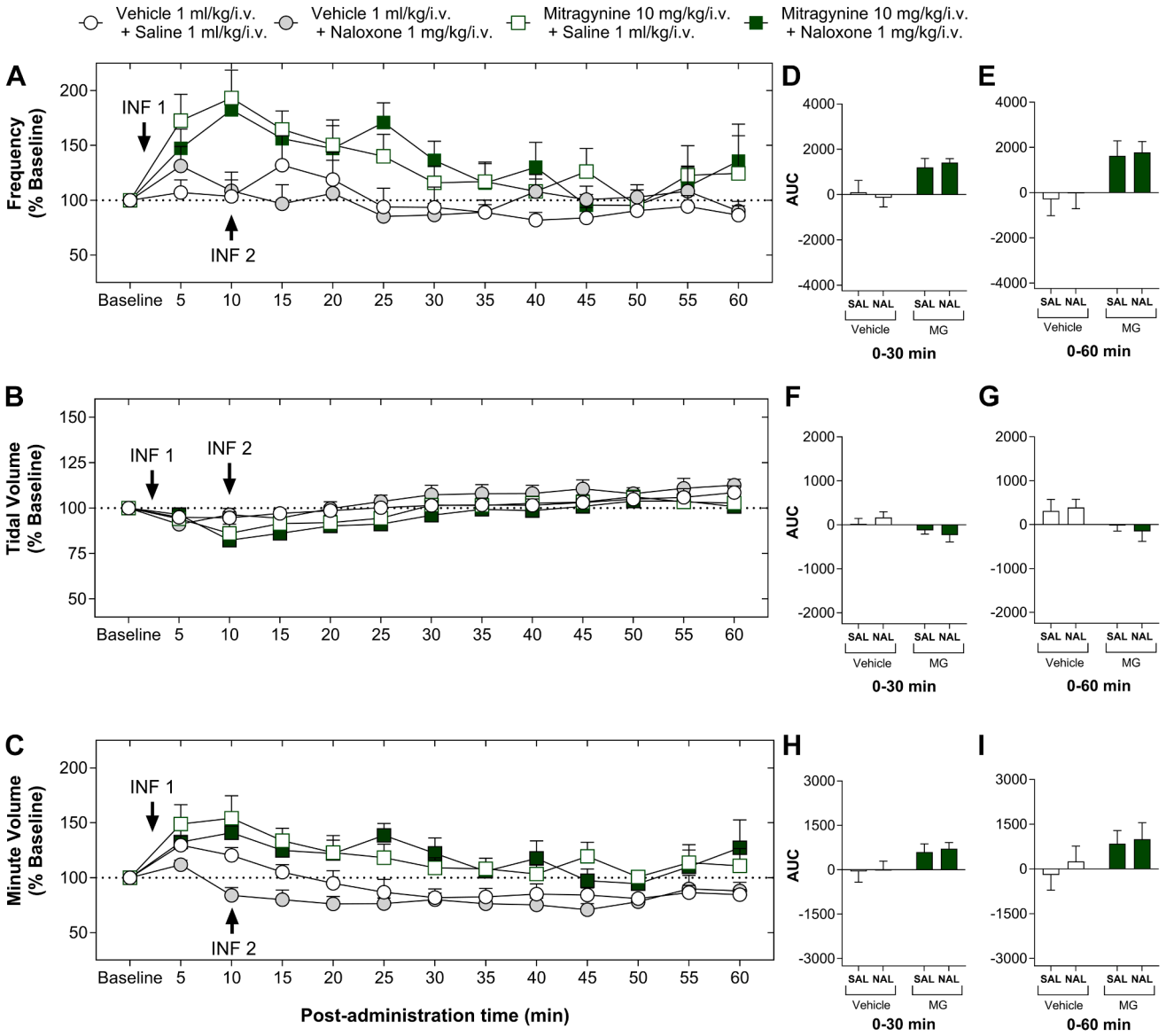


Fig. 5. Time course and AUC of MG and naloxone effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels) following intravenous administration of MG with naloxone rescue treatment in rats. MG was administered (10 mg/kg), followed by naloxone (1 mg/kg) rescue treatments after 10 minutes. Data points for the time course (A–C) represent the mean ± SEM ($n = 8$), recorded in 5-minute bins continuously for up to 60 minutes post-MG infusion. Data for AUC are shown as mean ± SEM ($n = 8$) for 30 (D, F, and H) and 60 (E, G, and I) minutes post-MG infusion. Statistically significant differences between treatments are indicated (P values).

For tidal volume, MG showed a near-significant effect at 30 minutes (Fig. 5F) ($F[1,7] = 5.26$; $P = .055$) and a significant effect at 60 minutes (Fig. 5G) ($F[1,7] = 7.18$; $P = .032$), with no significant naloxone effects or interactions. For minute volume, 2-way RM ANOVA showed no significant effects of MG, naloxone, or their interaction at either 30 (Fig. 5H) or 60 minutes (Fig. 5I). These results indicate that naloxone did not effectively antagonize the respiratory effects of MG, suggesting a mechanism distinct from typical opioid activity.

3.2.3. 7-HMG + naloxone

Naloxone effectively reversed the respiratory depression induced by 7-HMG. Time courses for respiratory frequency, tidal volume, and minute volume are presented in Fig. 6A–C, respectively. Two-way RM ANOVA for respiratory frequency at 30 minutes showed that while main effects of naloxone ($F[1,7] = 0.460$; $P = .519$) and 7-HMG ($F[1,7] = 0.0922$; $P = .770$) were not significant, a significant naloxone × 7-HMG interaction ($F[1,7] = 12.5$; $P = .010$) was observed (Fig. 6D). Further analysis of the interaction using multiple comparisons demonstrated that neither 7-HMG + saline nor 7-HMG + naloxone administration significantly impacted respiratory frequency compared with control. The direct comparison between 7-HMG + saline and 7-HMG + naloxone further revealed a significant difference ($P = .03$). At 60 minutes, no significant main effects or interactions were detected (Fig. 6E).

For tidal volume, 2-way RM ANOVA demonstrated significant effects of naloxone ($F[1,7] = 18.0$; $P = .004$), 7-HMG ($F[1,7] = 10.4$; $P = .014$), and naloxone × 7-HMG interaction ($F[1,7] = 11.7$; $P = .011$) at 30 minutes (Fig. 6F). Despite a significant main effect of naloxone, there was no significant difference between vehicle + naloxone and saline + saline control. Post hoc analysis also revealed significant depression with 7-HMG + saline compared

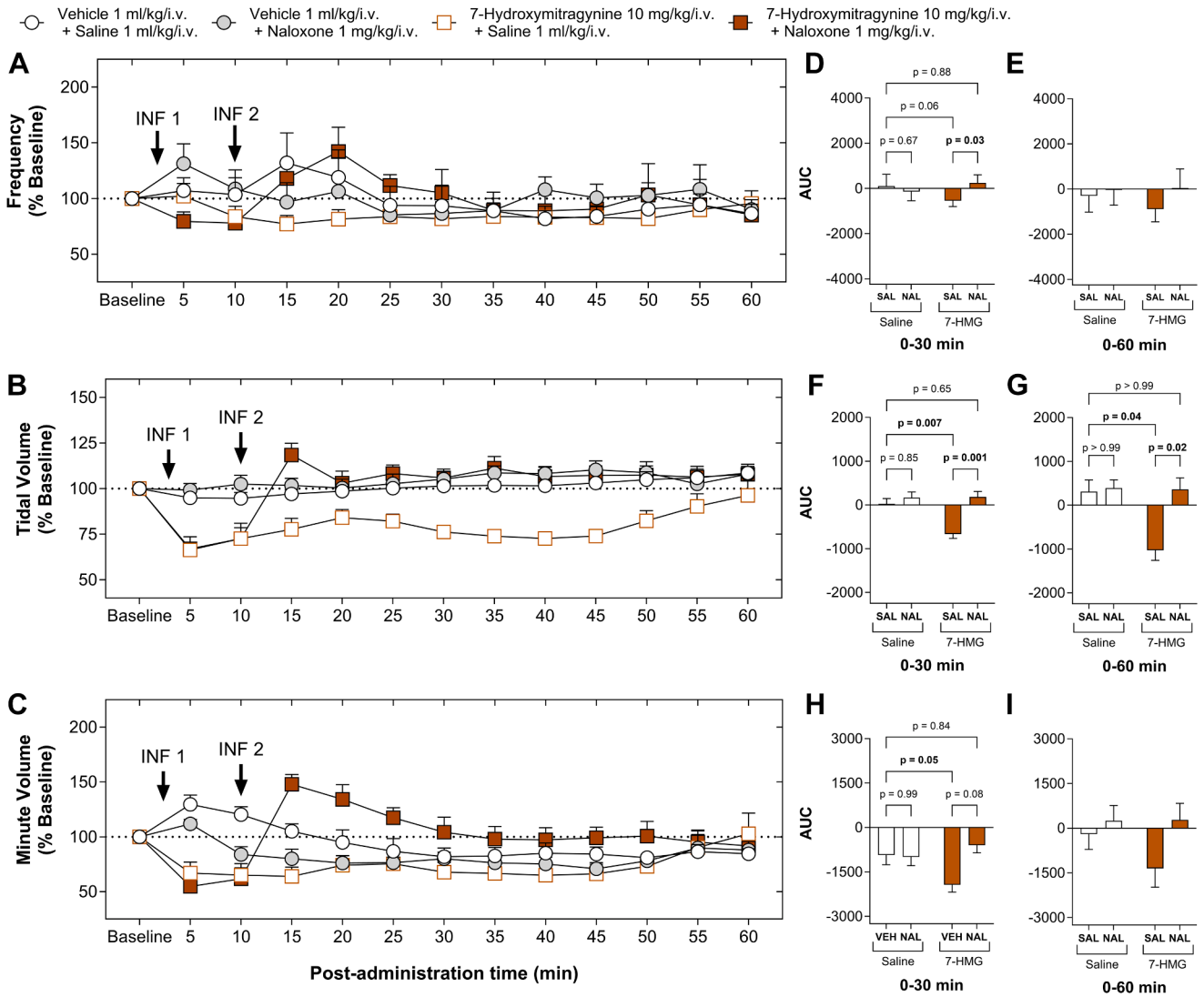


Fig. 6. Time course and AUC of 7-HMG and naloxone effects on respiratory parameters in rats. Respiratory frequency (upper panels), tidal volume (middle panels), and minute ventilation (lower panels) following intravenous administration of 7-HMG with naloxone rescue treatment in rats. 7-HMG was administered (10 mg/kg), followed by naloxone (1 mg/kg) rescue treatments after 10 minutes. Data points for the time course (A–C) represent the mean ± SEM ($n = 8$), recorded in 5-minute bins continuously for up to 60 minutes post-7-HMG infusion. Data for AUC are shown as mean ± SEM ($n = 8$) for 30 (D, F, and H) and 60 (E, G, and I) minutes post–7-HMG infusion. Statistically significant differences between treatments are indicated (P values).

with that with control ($P = .007$), which was not observed when naloxone was coadministered ($P = .65$). Additionally, the 7-HMG + saline and 7-HMG + naloxone treatments were significantly different from each other ($P = .001$). These effects on tidal volume persisted at 60 minutes for naloxone ($F[1,7] = 18.0$; $P = .004$), 7-HMG ($F[1,7] = 10.4$; $P = .014$), and naloxone × 7-HMG interaction ($F[1,7] = 11.7$; $P = .011$) (Fig. 6G). At 60 minutes, 7-HMG + saline showed significant depression compared with control ($P = .04$), whereas the 7-HMG + naloxone group was not significantly different from control. The difference between 7-HMG + saline and 7-HMG + naloxone treatments remained significant ($P = .02$).

For minute volume, despite nonsignificant main effects of naloxone ($F[1,7] = 3.85$; $P = .090$) and 7-HMG ($F[1,7] = 1.58$; $P = .249$) at 30 minutes, a significant naloxone × 7-HMG interaction ($F[1,7] = 26.7$; $P = .001$) was observed (Fig. 6H). Post hoc analysis at 30 minutes revealed a significant minute volume depression in the 7-HMG + saline group compared with that in the control group ($P = .05$), which was nonsignificant when naloxone was also administered. At 60 minutes, the interaction approached but did not reach significance (Fig. 6I) ($F[1,7] = 5.07$; $P = .059$).

4. Discussion

In the present study, both morphine and 7-HMG produced significant respiratory depression as indicated by decreases in breathing frequency, tidal volume, and minute volume, whereas MG unexpectedly increased respiratory frequency. Naloxone significantly reversed (ie, antagonized) the respiratory depressant effects of both morphine and 7-HMG, whereas the respiratory stimulant effects of MG were not antagonized or otherwise modified by naloxone administration. These results provide strong evidence that 7-HMG shares respiratory depressant effects with classical opioids, whereas MG increases respiration through mechanisms not involving opioid receptors.

Morphine, our reference compound, exhibited clear respiratory depression evidenced by decreases in tidal volume and minute

volume. At the highest dose tested (32 mg/kg), morphine significantly reduced tidal volume and minute volume, whereas no significant effects were observed at the lower dose (10 mg/kg), indicating dose-dependent respiratory depression under ambient air conditions. These effects were completely reversed by naloxone with an overshoot in respiration, confirming the well-established role of opioid receptor mechanisms in morphine-induced respiratory depression.[49–51] Rebound hyperventilation, or an overshoot in respiration, can occur following administration of naloxone after morphine-induced respiratory depression. Administration of naloxone rapidly displaces morphine from MORs in the brainstem, abruptly restoring excitatory input to respiratory centers and together with compensatory physiological mechanisms, triggering the overshoot observed.[2,3,52,53] Our findings are consistent with previously documented morphine-induced decreases in respiration across multiple species using various routes of administration.[1,54–56]

In the current study, MG did not produce respiratory effects typical of most opioid receptor agonists, that is, did not decrease respiration up to 17.8 mg/kg. Doses of MG larger than 17.8 mg/kg produced overt signs of toxicity and were not fully tested. Instead, MG produced significant increases in respiratory frequency. In previously published studies, MG was reported to decrease some respiratory parameters.[33] Lyophilized kratom tea also decreased respiration rates.[57] However, the experimental parameters of the previous studies (eg, oral administration in mice) differed from the experimental parameters of the current study (eg, intravenous administration in rats). With respect to route of administration, cytochrome 450-mediated conversion of MG into its active metabolites is expected to be greater following oral as compared with intravenous administration.[58–61] Although metabolism to sufficiently large amounts of 7-HMG could potentially contribute to the effects produced by MG administration, our group has demonstrated MG-induced antinociception in mice is evident when the amounts of 7-HMG measured in the brain are far less than the amounts required for antinociceptive effects.[62] Therefore, experimental support for a MG prodrug hypothesis is currently lacking, and any differences in 7-HMG formation across routes of administration do not appear to explain variable outcomes in studies of antinociception. Alternatively, the effects of MG appear to differ consistently between mice[33,57] and rats.[32] For example, under experimental conditions in rats that are highly sensitive to the antinociceptive effects of 7-HMG and other opioid agonists, our group has demonstrated that MG administration is devoid of antinociceptive effects.[24] The present study extends these findings to include respiratory effects, which in rats are not merely lacking but rather opposite in direction to the respiratory depressant effects of opioids. Intravenous administration of MG produced no significant change in temperature up to 10 mg/kg,[63] indicating that the MG-induced increase in frequency may not be due to an increase in the metabolic rate. The MG-induced increase in frequency might be due to its in vitro α2-adrenergic antagonistic activity,[64] which would prevent norepinephrine negative feedback,[65,66] resulting in increased alertness, arousal, and overall activity.

Results for 7-HMG demonstrated significant dose-dependent respiratory depression, particularly at the 10-mg/kg dose. Importantly, we observed distinct time-dependent effects in the respiratory parameters. At 30 minutes postadministration, 7-HMG significantly decreased respiratory frequency and minute volume at both 3.2- and 10-mg/kg doses, but these effects were not maintained at 60 minutes. This temporal profile indicates that 7-HMG's respiratory depressant effects peak earlier and begin to diminish over time, contrasting with morphine, which maintained effects across both 30 and 60 minutes. The reductions in respiratory frequency, tidal volume, and minute volume by 7-HMG observed in this study parallel the characteristic respiratory profile of classical μ-opioid agonists such as morphine, which are consistent with prior findings where 7-HMG showed respiratory depressant effects in mice.[33] Notably, the reversal of 7-HMG-induced respiratory depression by naloxone provides pharmacological evidence for opioid receptor-mediated effects. The efficacy of naloxone antagonism, particularly evident in tidal volume measurements, supports substantial MOR involvement in 7-HMG's respiratory effects. The ED_{50} of 7-HMG (6.98 mg/kg) to decrease minute volume was 4.5-fold more potent than morphine (31.5 mg/kg). These findings align with previous in vivo characterization, demonstrating that 7-HMG has 3- to 13-fold higher potency than morphine.[24–26] Collectively, these results suggest that 7-HMG functions primarily as an MOR agonist.

The opposing respiratory effects of MG and 7-HMG suggest that MG—the most abundant kratom alkaloid, constituting up to 66% of the total alkaloidal fraction[67]—may counteract the respiratory depressive effects of 7-HMG. In the study by Hill et al,[33] MG was shown to reduce minute volume; however, its respiratory depressant effects exhibited a ceiling effect, such that respiratory depression did not continue to worsen at higher doses of MG. In contrast, 7-HMG did not display a ceiling effect.[33] Furthermore, Wilson et al[57] demonstrated that although lyophilized kratom tea (containing all alkaloids in the plant) decreased respiratory rate, the reduction in respiration caused by morphine was significantly greater than that produced by the kratom tea. These findings suggest that MG could mitigate the respiratory depressant effects of 7-HMG, given that the magnitude of 7-HMG-induced respiratory depression is similar to that of morphine. MG is metabolized to 7-HMG in liver microsomes, with this conversion occurring more efficiently in human liver microsomes than in mouse liver microsomes.[68,69] However, the extent to which 7-HMG contributes to the respiratory depressant effects of MG may be limited because MG produces a ceiling effect on respiration.[33] Furthermore, MG-induced antinociception in mice is observed even when brain concentrations of 7-HMG are significantly lower than those required to elicit antinociceptive effects on their own.[62] In rats, ketoconazole was shown to inhibit the metabolism of MG to 7-HMG, indicating that rat liver microsomes also mediate this metabolic pathway.[59] In humans, co-administration of itraconazole with kratom tea reduced 7-HMG exposure by 56%, suggesting that 7-HMG exposure following kratom consumption may be higher in humans than in rodents.[70] Taken together, these results support the idea that MG contributes to the improved safety profile of kratom. However, caution is warranted with higher doses of MG because administration of a high dose of MG (17.8 mg/kg) induced seizure-like activity without causing respiratory depression. This observation aligns with poison control center data (2011–2017), documenting 113 seizures among 1174 kratom exposures.[71]

Adding to these concerns is the rapid proliferation of kratom-associated products specifically marketed for their high 7-HMG content.[27,28] An online market analysis conducted in February 2025 identified 304 semisynthetic kratom products. The majority (82.2%) contained only 7-HMG and were formulated as chewable or sublingual tablets, shots, or gummies. Products containing 7-HMG in combination with another semisynthetic compound, mitragynine pseudoindoxyl, accounted for 14.5% and were most commonly marketed as kratom.[72] Such formulations may be mistakenly perceived as typical kratom products, increasing the risk of unintentional exposure to pharmacologically significant doses of 7-HMG. The ability of 7-HMG to induce respiratory depression similar to morphine, highlights the urgent need for the regulation of these 7-HMG–containing products.

Although the findings of this study contribute significantly to our understanding of the respiratory effects of MG and 7-HMG and their interaction with naloxone, several limitations should be considered. First, we examined respiration after acute administration only; repeated administration might yield different respiratory profiles. Second, although we focused on respiratory parameters, a comprehensive safety evaluation would require assessment of additional physiological systems, such as cardiovascular and carcinogenic toxicity. Third, kratom contains numerous alkaloids; additional research is needed to determine the respiratory effects of other constituent alkaloids and their combinations. Future studies are also needed to investigate the molecular mechanisms underlying the distinct respiratory profile of MG, which was not affected by naloxone.

5. Conclusions

These respiratory depression findings have direct implications for evaluating kratom's safety profile and informing regulatory decisions. The observation that MG does not produce respiratory depression, and may even stimulate respiration, is particularly intriguing. However, the potent respiratory depressant effects of 7-HMG raise significant safety concerns, especially considering the increasing presence of novel semisynthetic products that may contain up to 98% 7-HMG.[28,72] Importantly, the respiratory depression produced by 7-HMG is sensitive to naloxone, which may have lifesaving implications in emergency medicine.

Abbreviations

AUC, area under the curve; 7-HMG, 7-hydroxymitragynine; MG, mitragynine; MOR, μ-opioid receptor; OIRD, opioid-induced respiratory depression; RM, repeated-measures.

Acknowledgments

The authors thank Dr Renata Marchette and Dr George Koob for providing training to J.L.W. on the use of rat whole body plethysmography. The authors also gratefully acknowledge Dr Aiden Hampson for his valuable input during the development of this project.

Financial support

This work was supported by National Institutes of Health National Institute on Drug Abuse [Grant R01 DA48353 to C.R.M. and L. R.M.] and [Grant UG3/UH3 DA048353 01 to C.R.M. and L.R.M.] and by Texas Tech University Health Sciences Center Office of Research and the Jerry H. Hodge School of Pharmacy.

Conflict of interest

The authors declare no conflicts of interest.

Data availability

The data that support the findings of this study are available within the article. Any additional information is available on request from the corresponding authors.

CRediT authorship contribution statement

Julio Zuarth Gonzalez: Conceptualization, Formal analysis, Investigation, Writing – Original Draft, Writing – Review and Editing. **Alexandria Ragsdale**: Investigation, Writing – Review and Editing. **Sushobhan Mukhopadhyay**: Resources, Writing – Review and Editing. **Christopher McCurdy**: Conceptualization, Funding acquisition, Resources, Supervision and Project administration, Writing – Original Draft, Writing – Review and Editing. **Lance McMahon**: Conceptualization, Funding acquisition, Supervision and Project administration, Writing – Original Draft, Writing – Review and Editing. **Samuel Obeng**: Formal analysis, Writing – Original Draft, Writing – Review and Editing. **Jenny Wilkerson**: Conceptualization, Formal analysis, Supervision and Project administration, Writing – Original Draft, Writing – Review and Editing.

Supplemental material

This article has supplemental material available at jpet.aspetjournals.org.

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